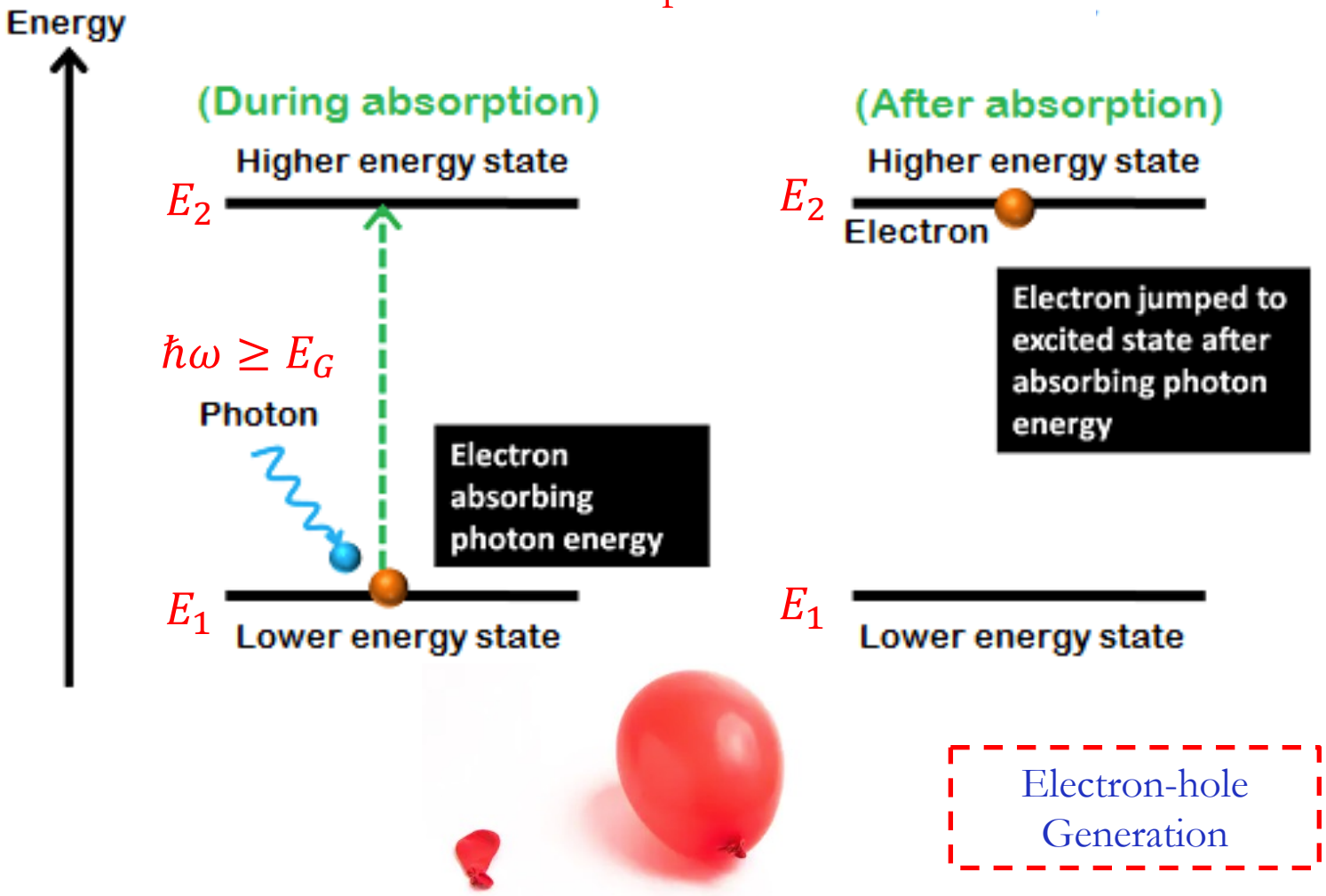
Semiconductors Fundamentals

The following subjects will be covered

- Semiconductors in General.
- The PN Junction.
- The Energy Band Diagram.
- Carrier Statistics in Semiconductors.
- PN Junction Band Diagram.
- Photon Generation and Absorption.
- Direct and Indirect Gap Materials.
- Examples.

Photon-matter Interactions

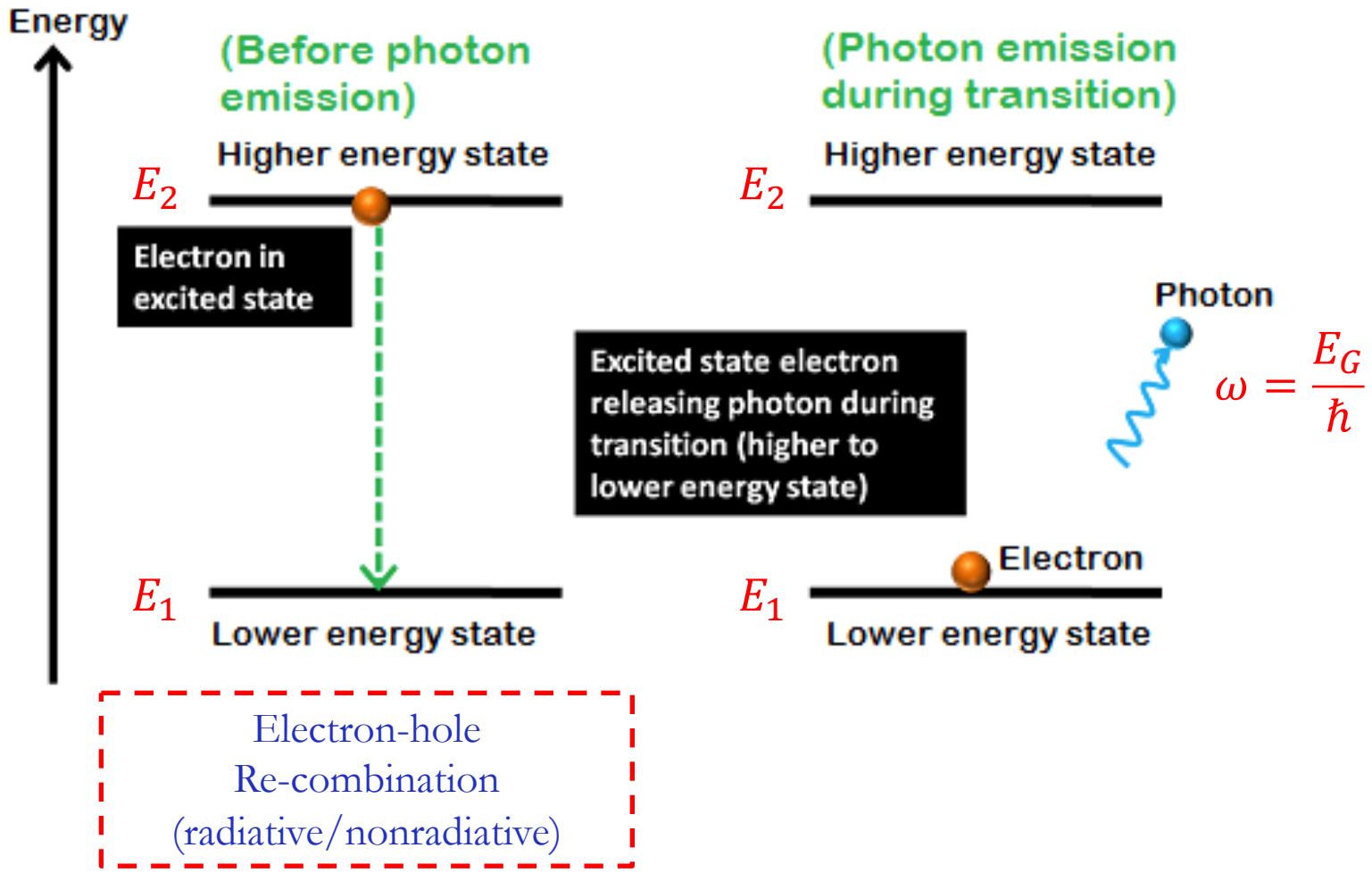
Absorption



Electron-hole Generation

Photon-matter Interactions

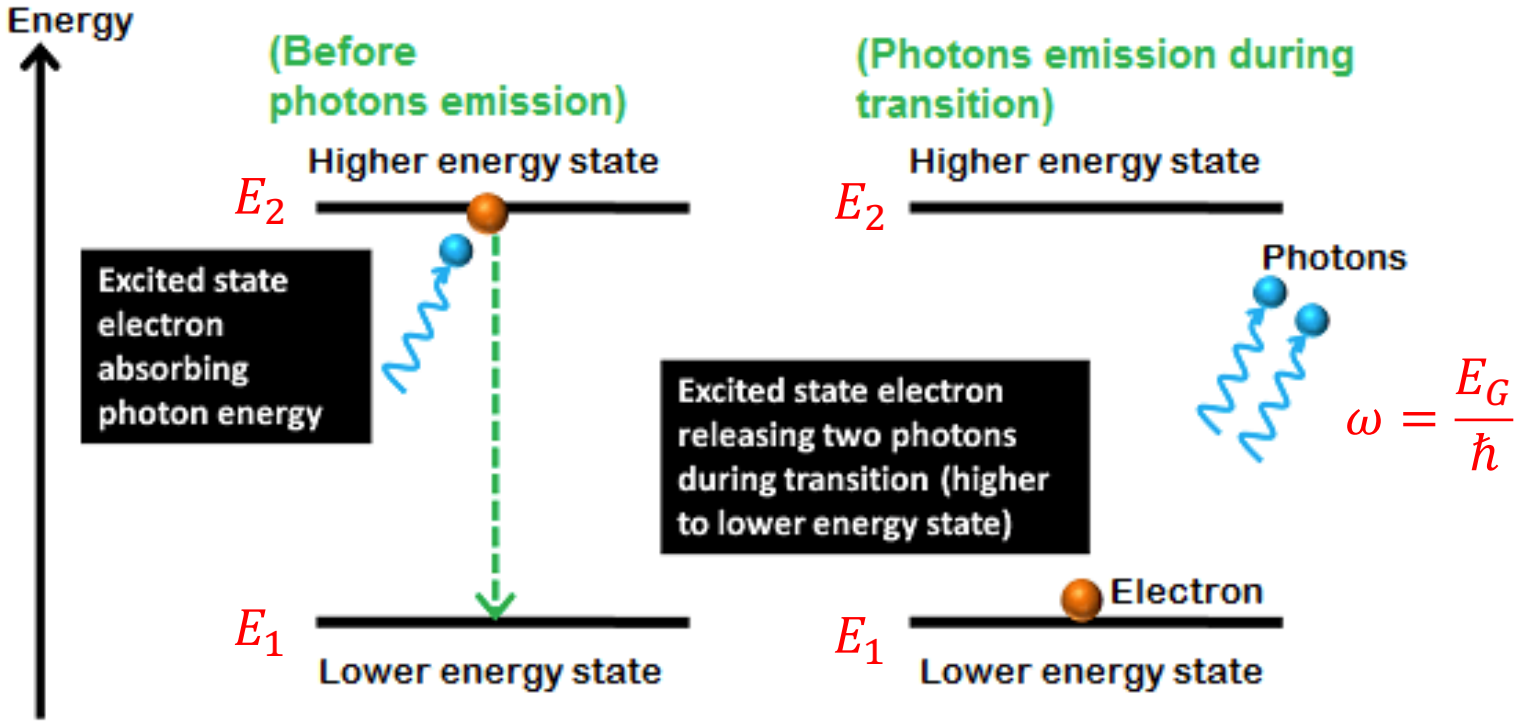
Spontaneous Emission



Electron-hole
Re-combination
(radiative/nonradiative)

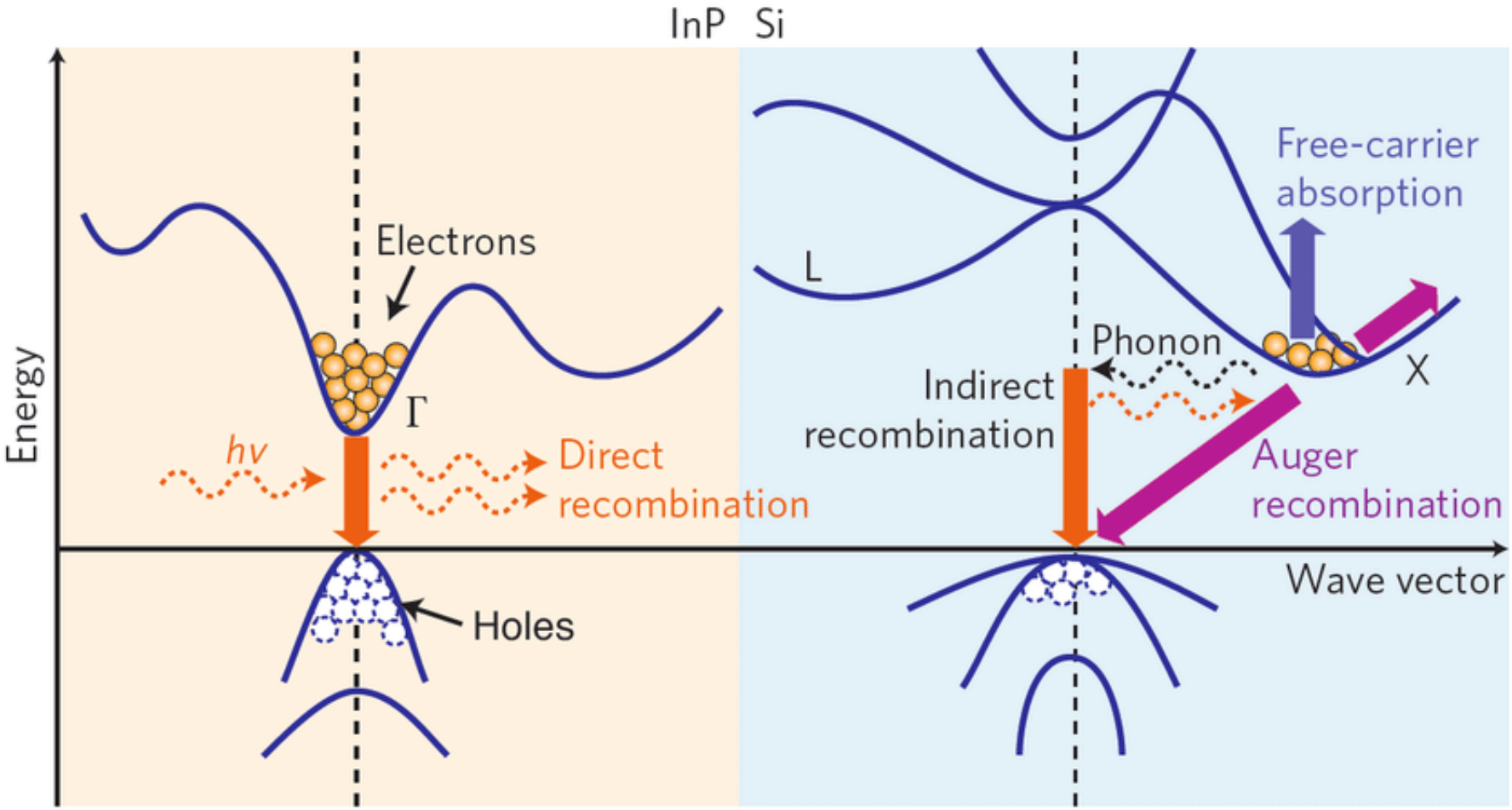
Photon-matter Interactions

Stimulated Emission



Electron-hole
Re-combination
(radiative/nonradiative)

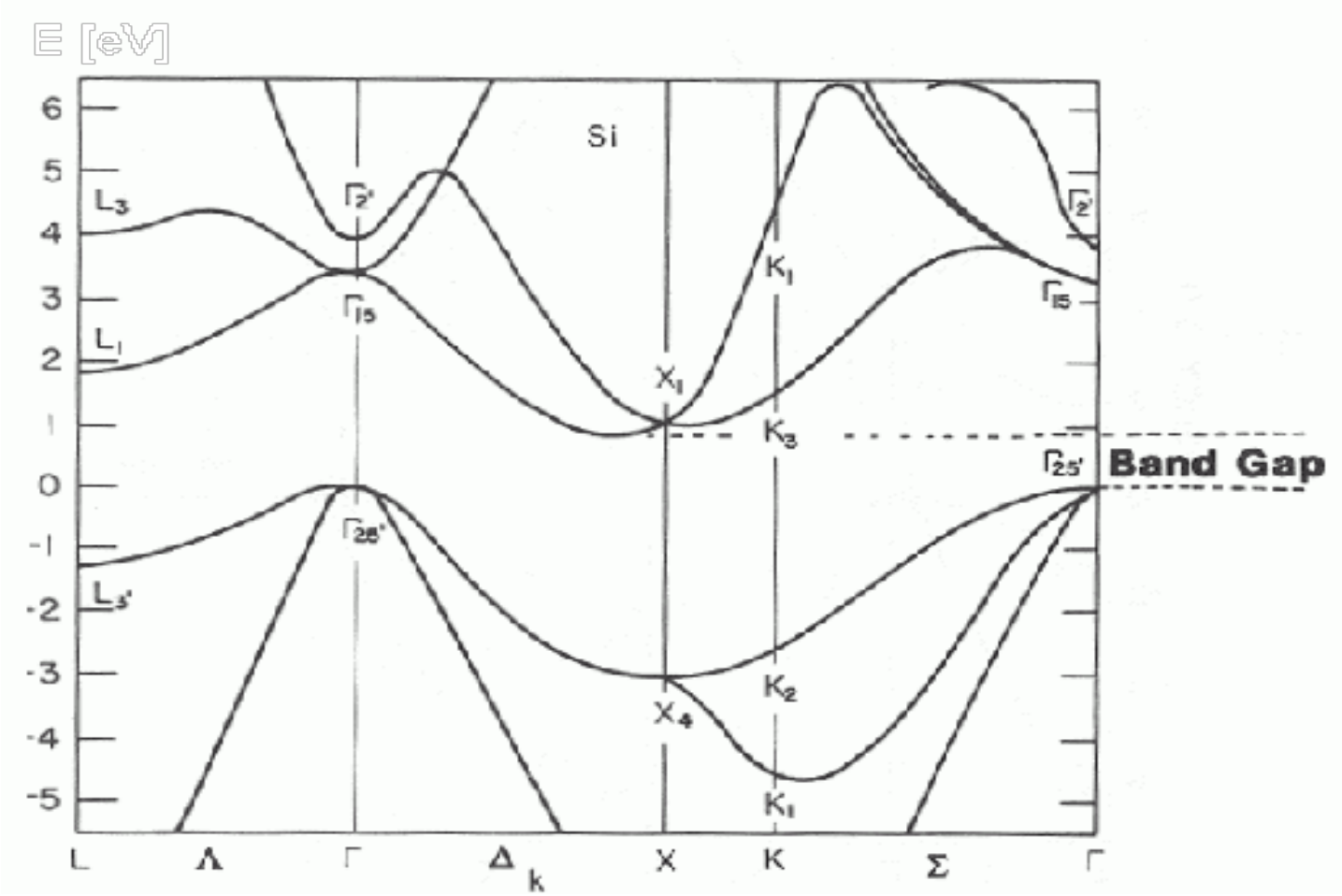
Direct and Indirect Gap Materials



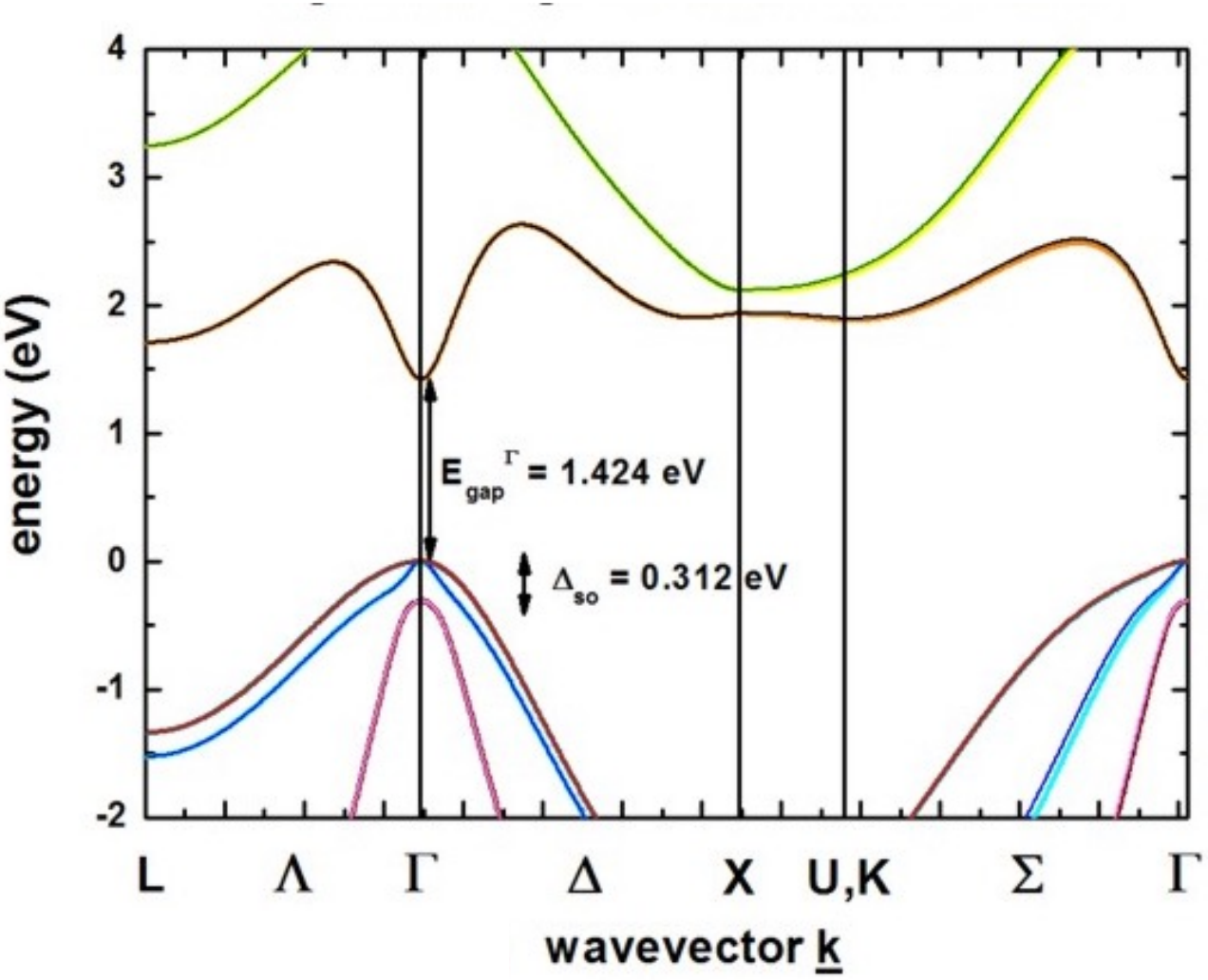
Direct and Indirect Gap Materials

	Material	Direct / IndirectBandgap	Band Gap Energy at 300 K (eV)
Elements	C (di	Indirect	5.47
	Ge	Indirect	0.66
	Si	Indirect	1.12
	Sn	Direct	0.08
Groups III-V compounds	GaAs	Direct	1.42
	InAs	Direct	0.36
	InSb	Direct	0.17
	GaP	Indirect	2.26
	GaN	Direct	3.36
	InN	Direct	0.70
	InP	Direct	1.344
Groups IV-IV compounds	α-SiC	Indirect	2.99
Groups II-VI compounds	ZnO	Direct	3.35
	CdSe	Direct	1.70
	ZnS	Direct	3.68

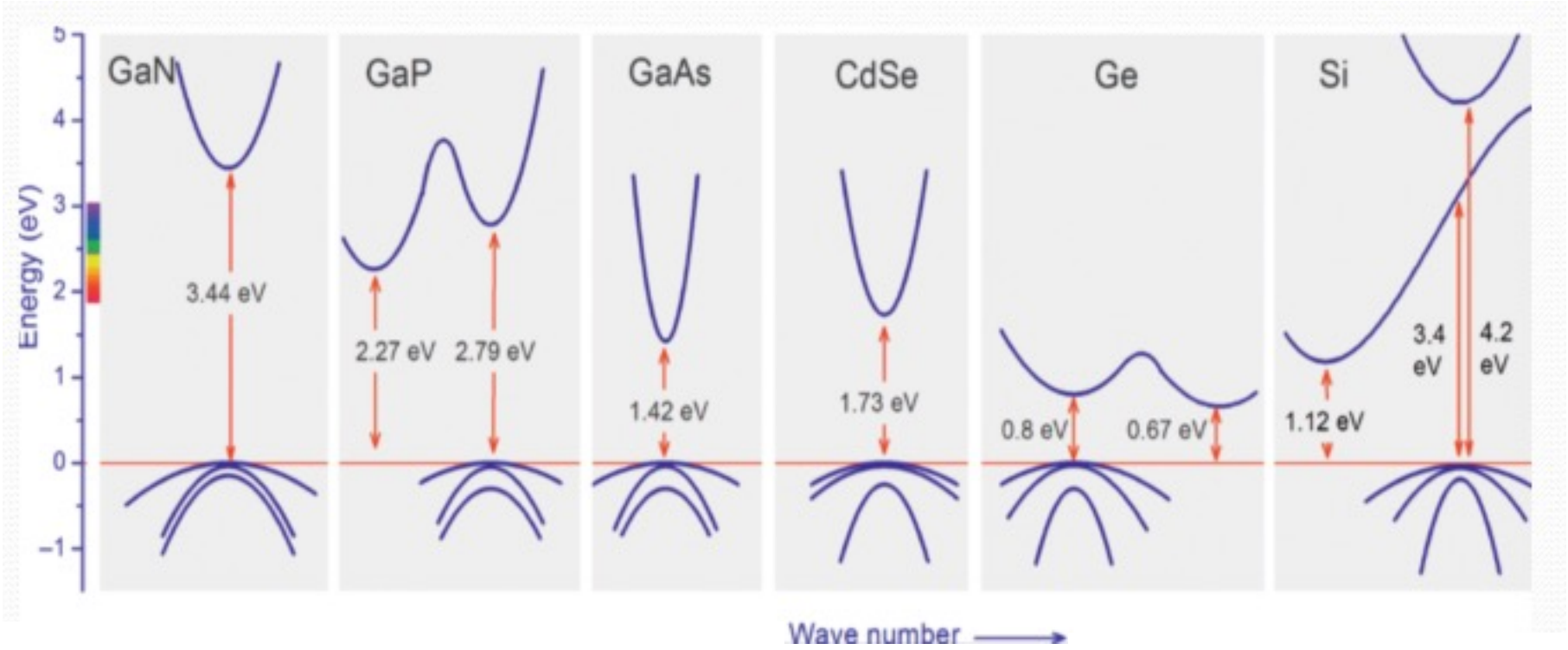
Si: Indirect Gap Material



GaAs: Direct Gap Material



GaAs: Direct Gap Material



Bandgap Engineering

Example: Emission in GaAs, GaN and InN.

$$\omega = \frac{E_G}{\hbar}$$

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega}$$

$$\lambda = \frac{2\pi c\hbar}{E_G}$$

$$\lambda_g = \frac{hc}{E_G}$$

GaAs band gap energy E_G=1.42 eV

$$\lambda_g = \frac{2\pi(3\times10^8)(1.055\times10^{-34})}{(1.42)(1.6\times10^{-19})}$$

$\lambda_g = 875\ nm$ NIR

GaN band gap energy E_G=3.49 eV

$\lambda_g = 356\ nm$ UV

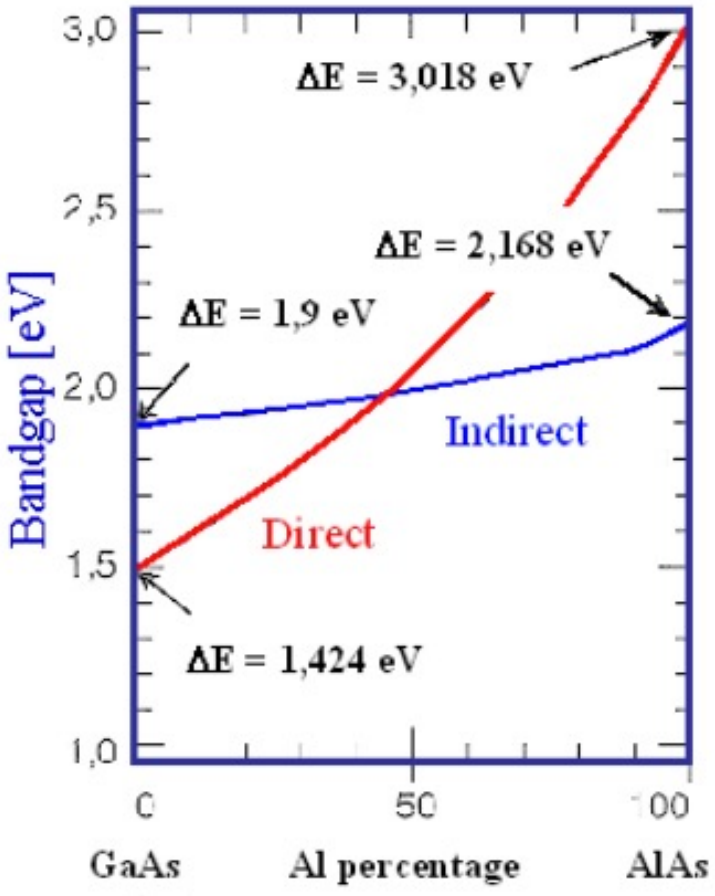
InN band gap energy E_G=0.69 eV

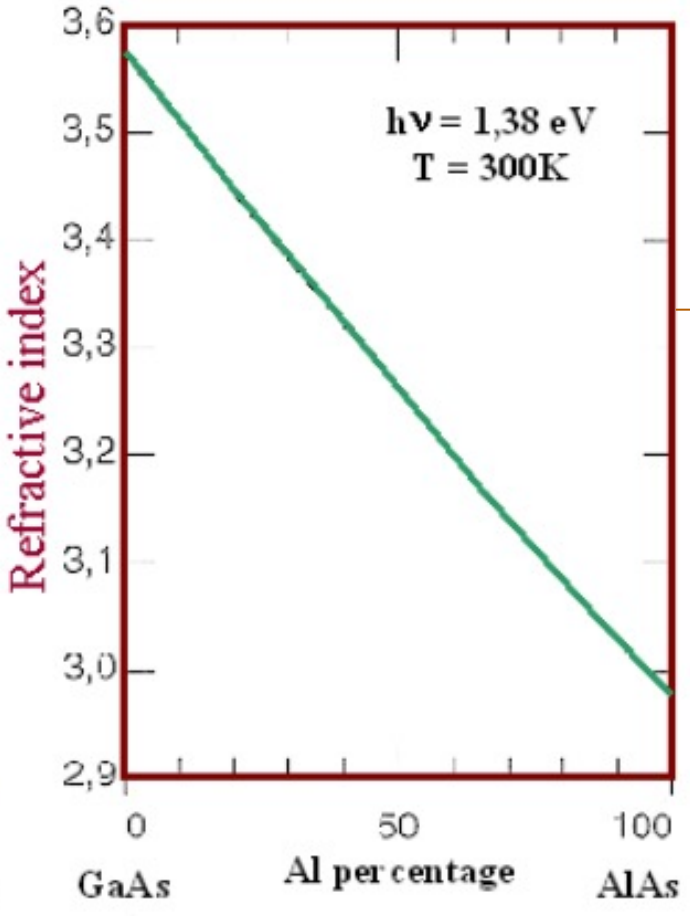
$\lambda_g = 1800\ nm$ IR

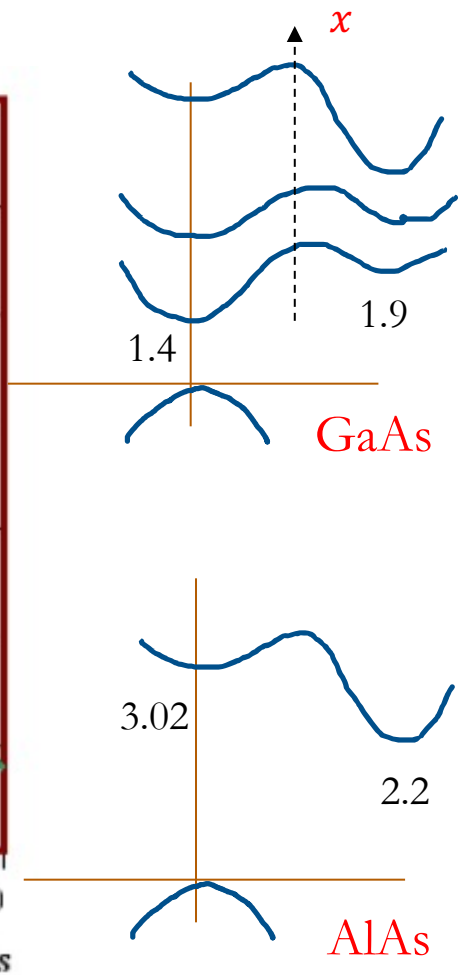
The alloy $In_xGa_{1-x}N$ is very useful for spanning the visible range.

Bandgap Engineering

Example: Emission in $Al_xGa_{1-x}As$.





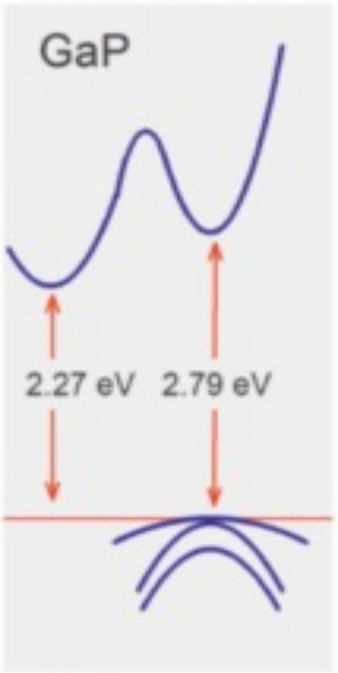


The Green Gap

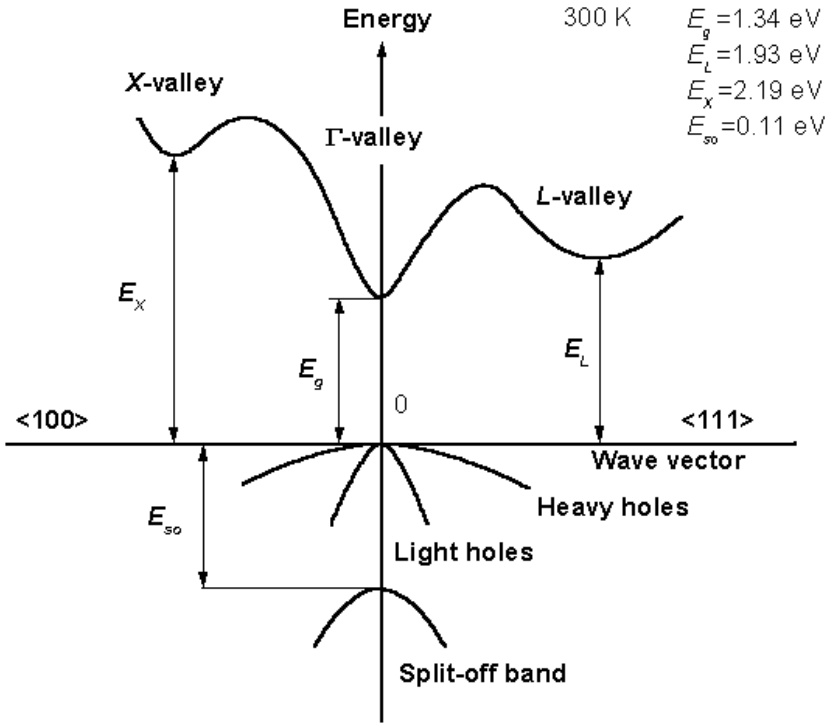
Green Band: 530-570 nm.

Green Band: 2.2-2.3 eV.

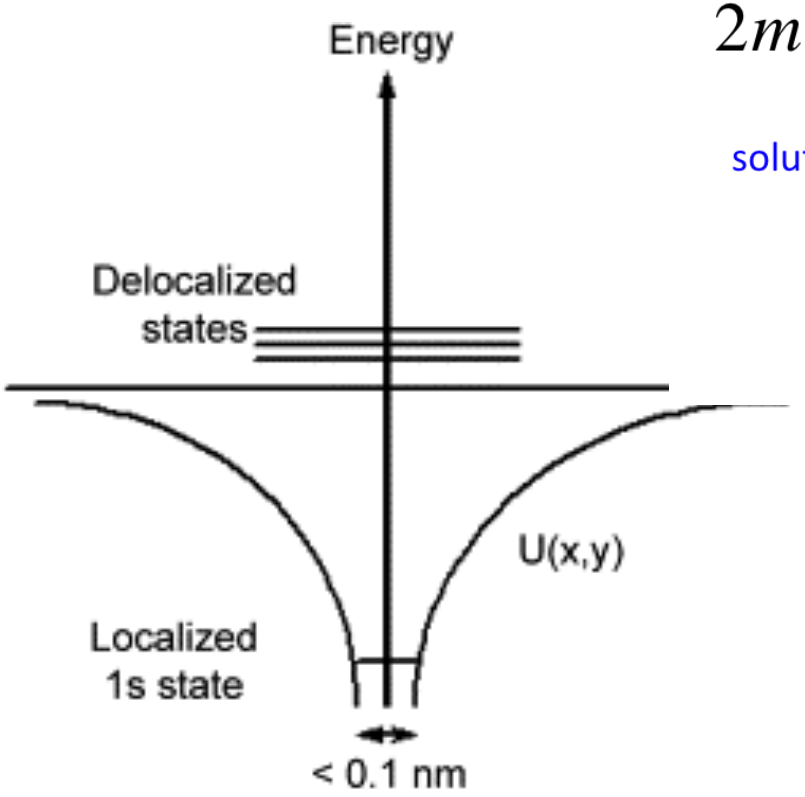
Emission in $Ga_xIn_{1-x}P$.



InP



Other Examples



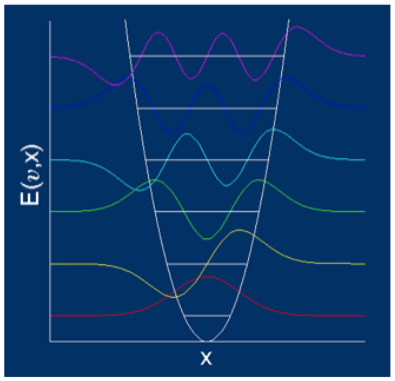
Harmonic Oscillator Schrödinger Equation

$$\frac{-\hbar}{2m}\frac{d^2\Psi}{dx^2}+\frac{kx^2}{2}\Psi=E\Psi$$

solution gives:

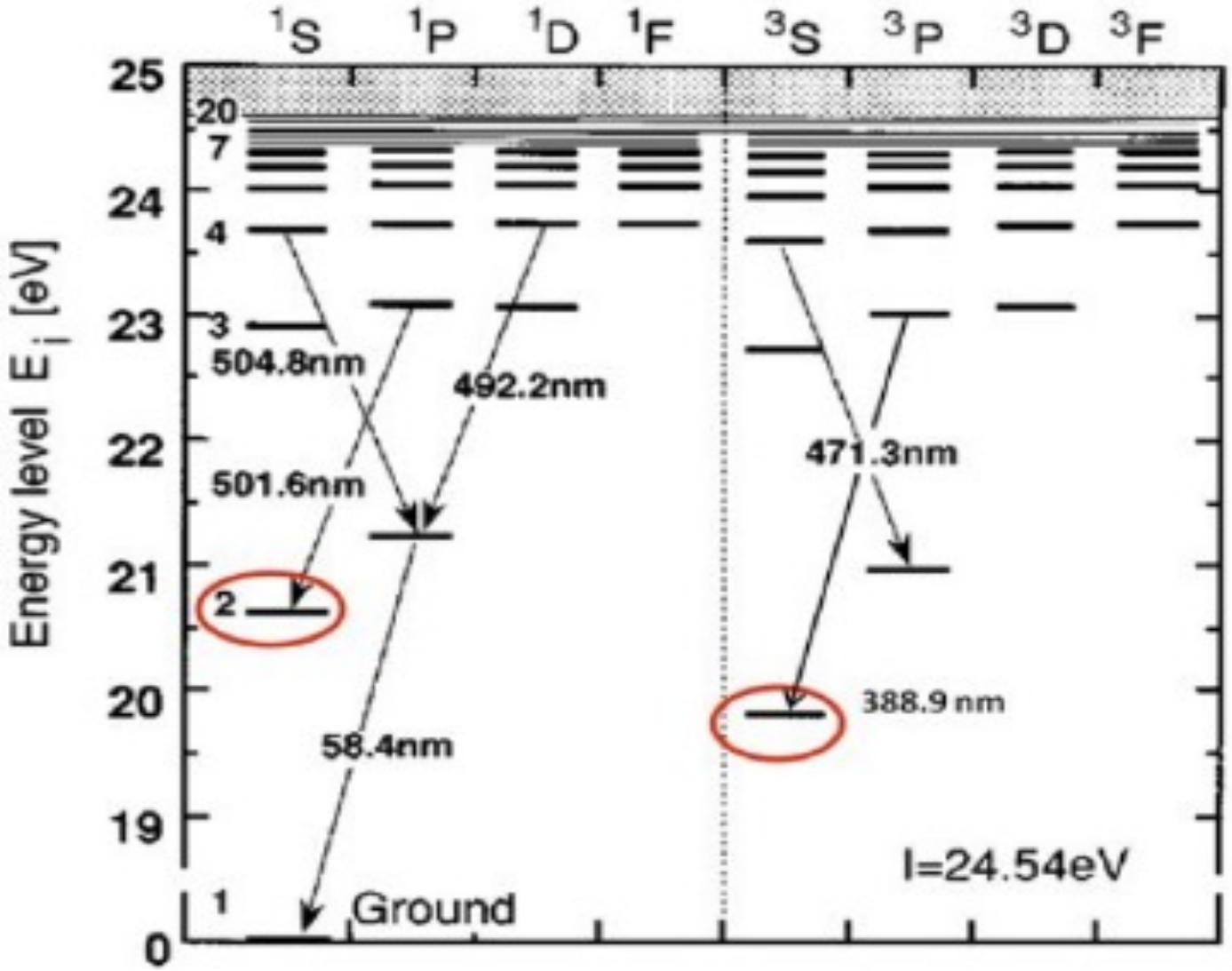
$$E_n=(n+\tfrac{1}{2})\hbar\omega$$

n = 0, 1, 2, 3 ...

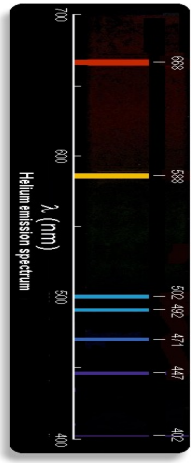


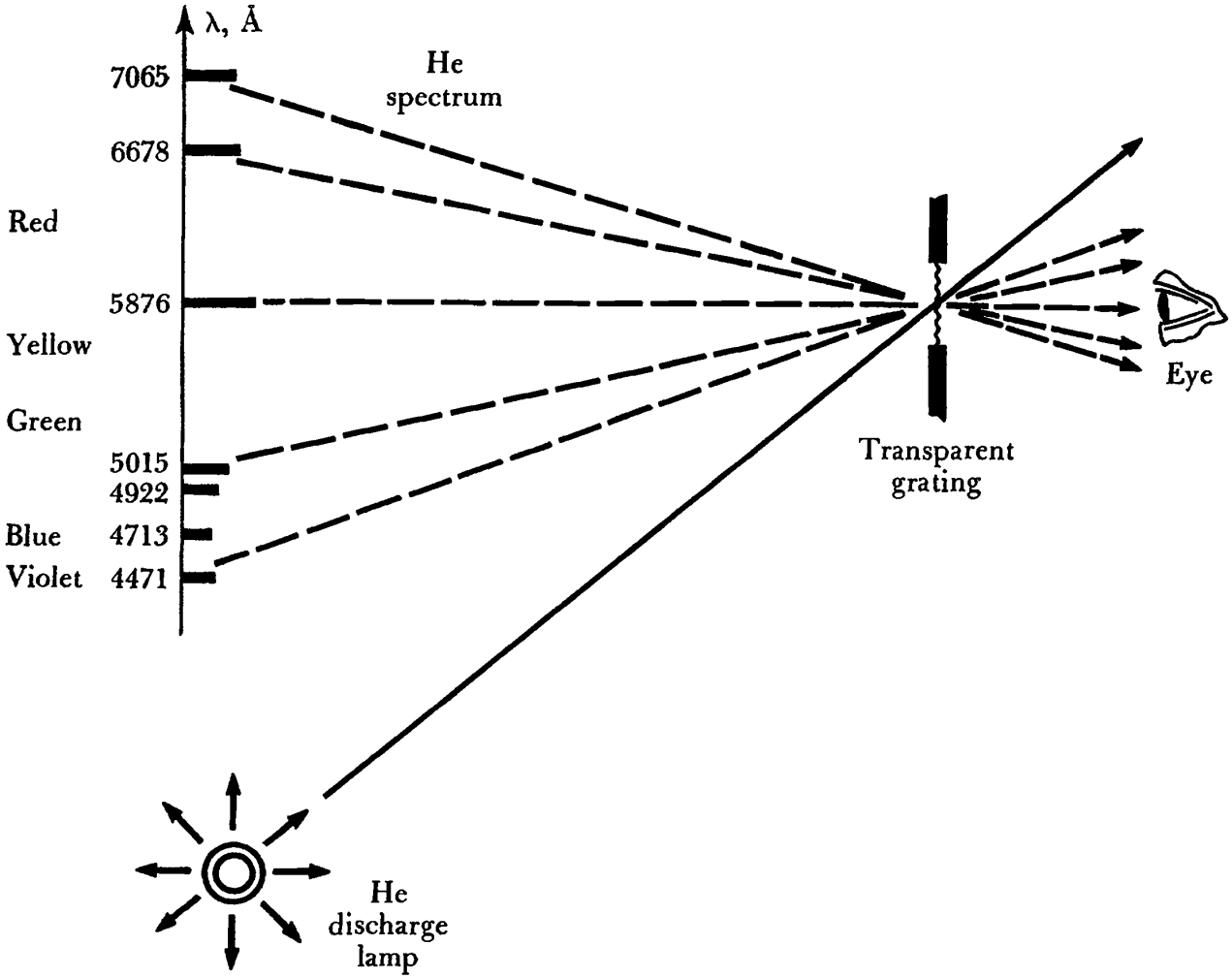
plot Ψ vs x
for different E_n values

Examples: Helium Atom

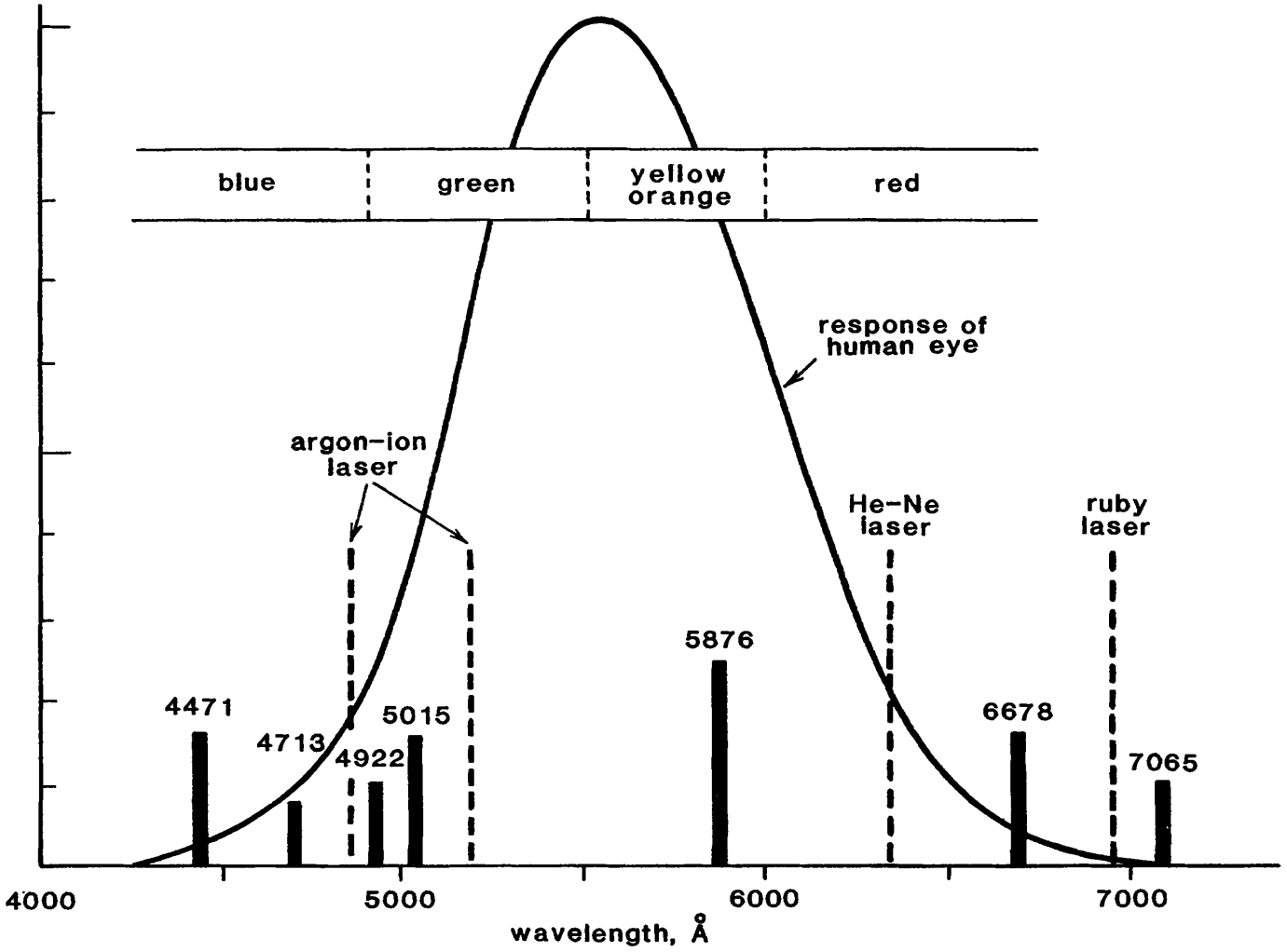


Examples: Helium Atom



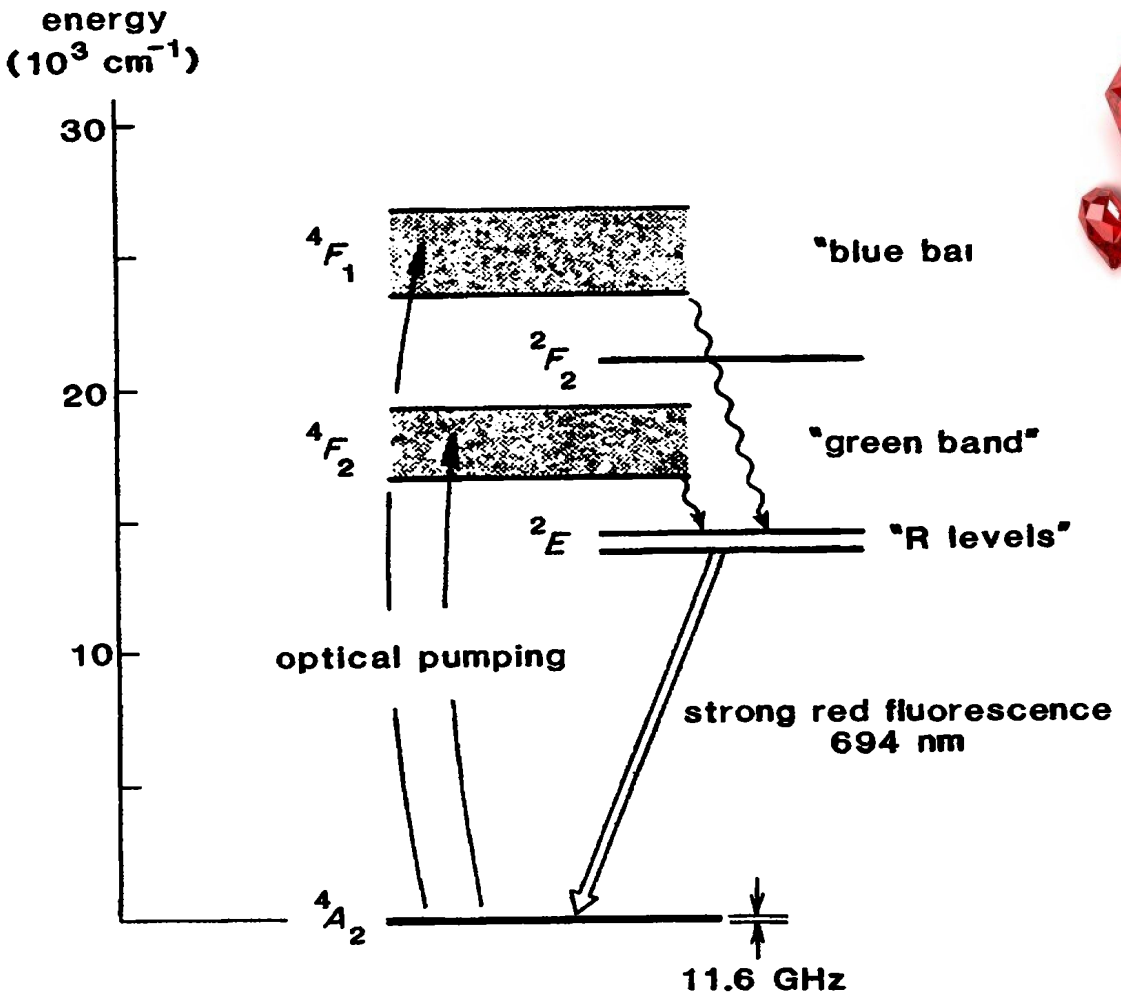


Examples: Helium Atom



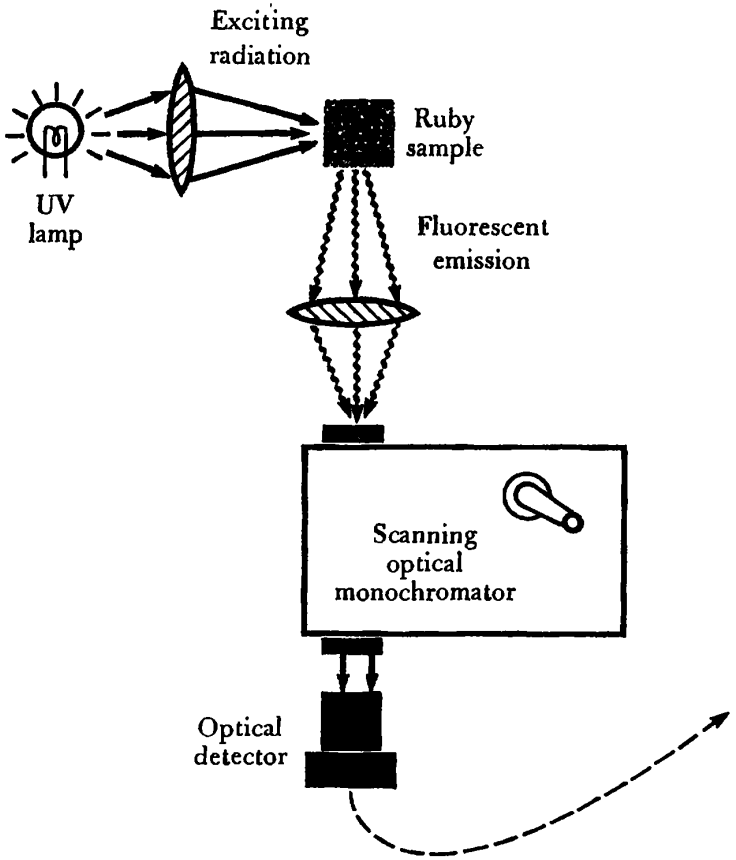
Examples: Ruby

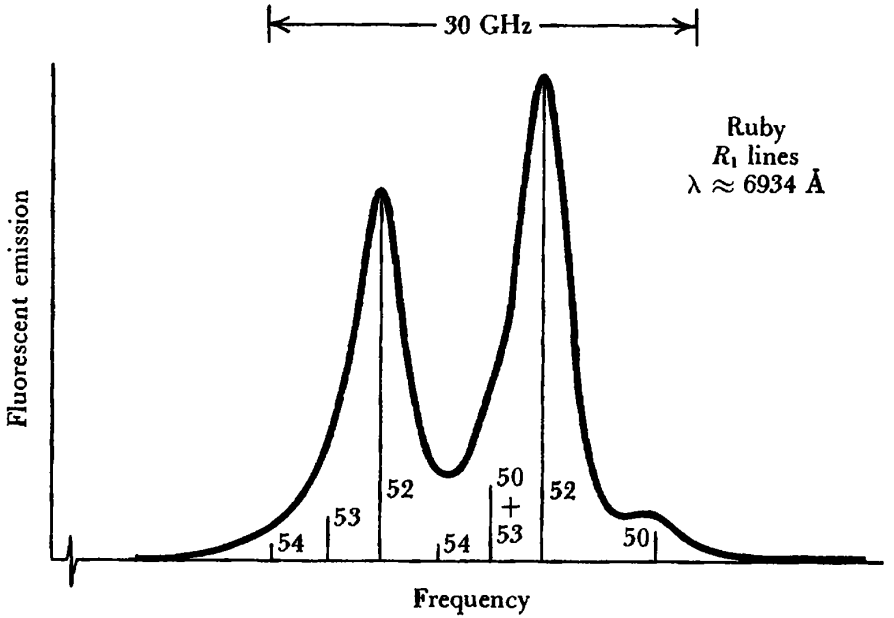






Examples: Ruby



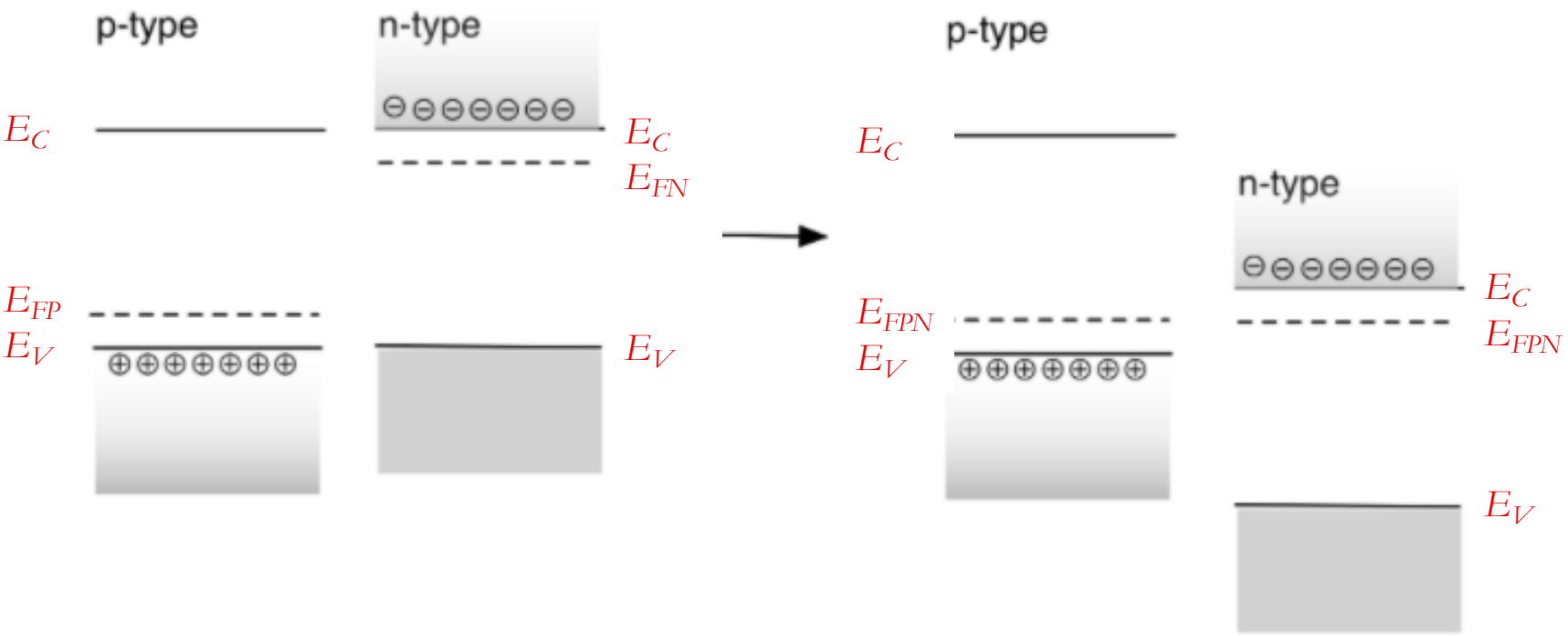


Light Emitting Diodes

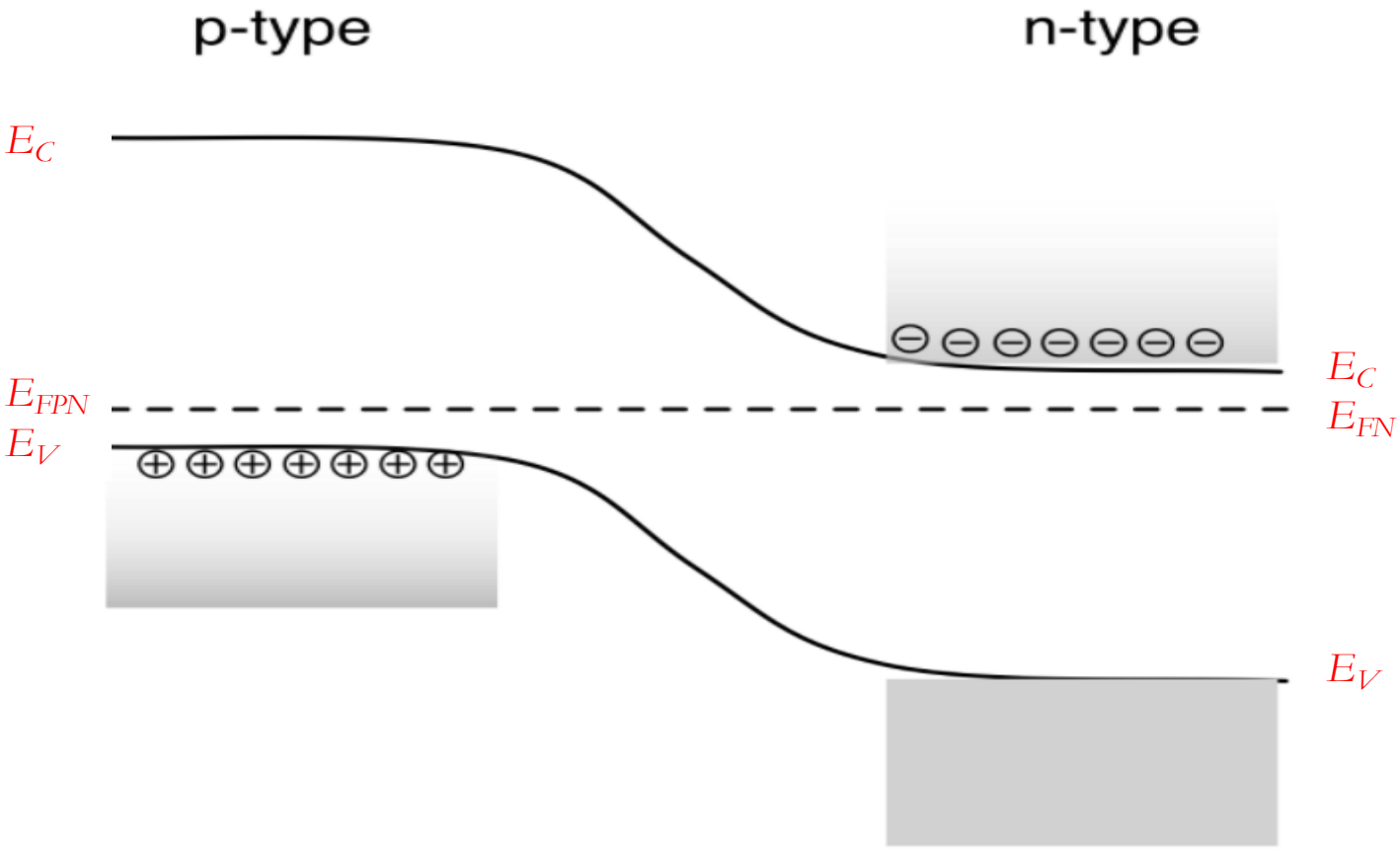
The following subjects will be covered

- The PN Junction Band Diagram
- LED Basic Design and Operation
- LED Efficiency

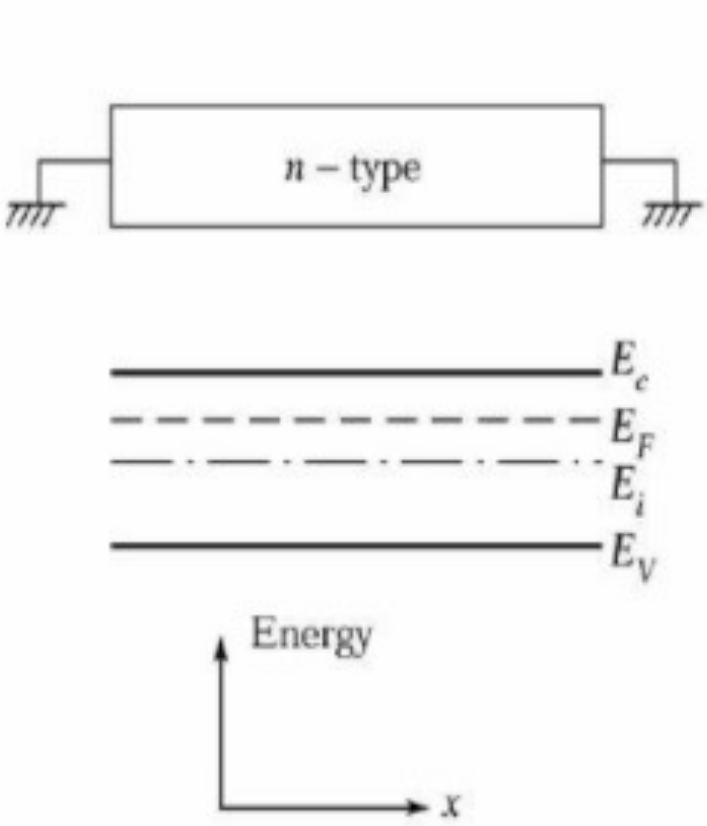
The PN Junction Band Diagram

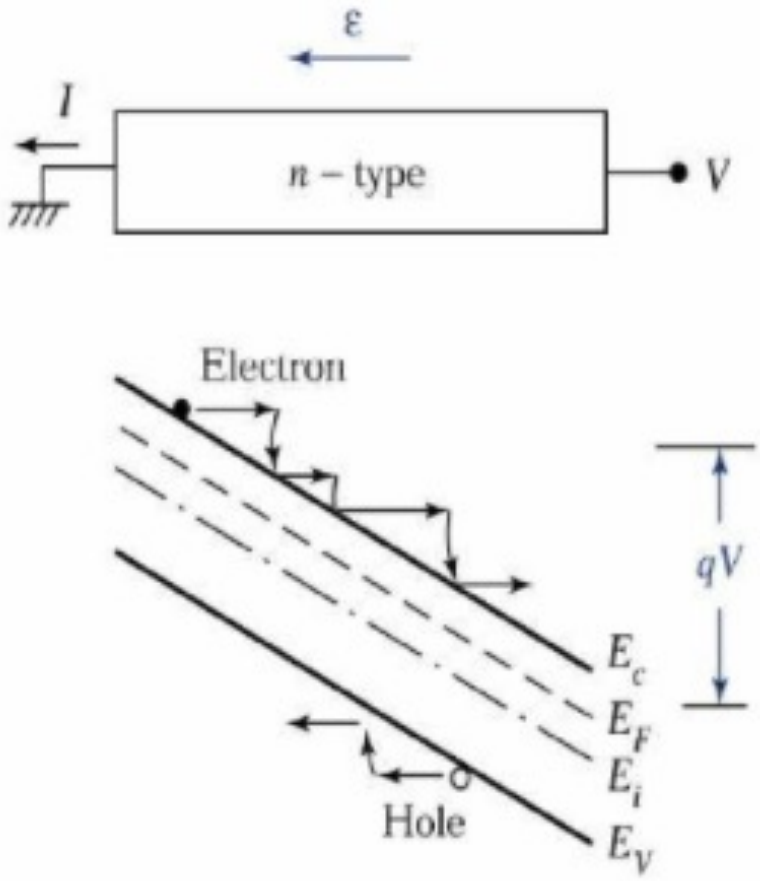


The PN Junction Band Diagram

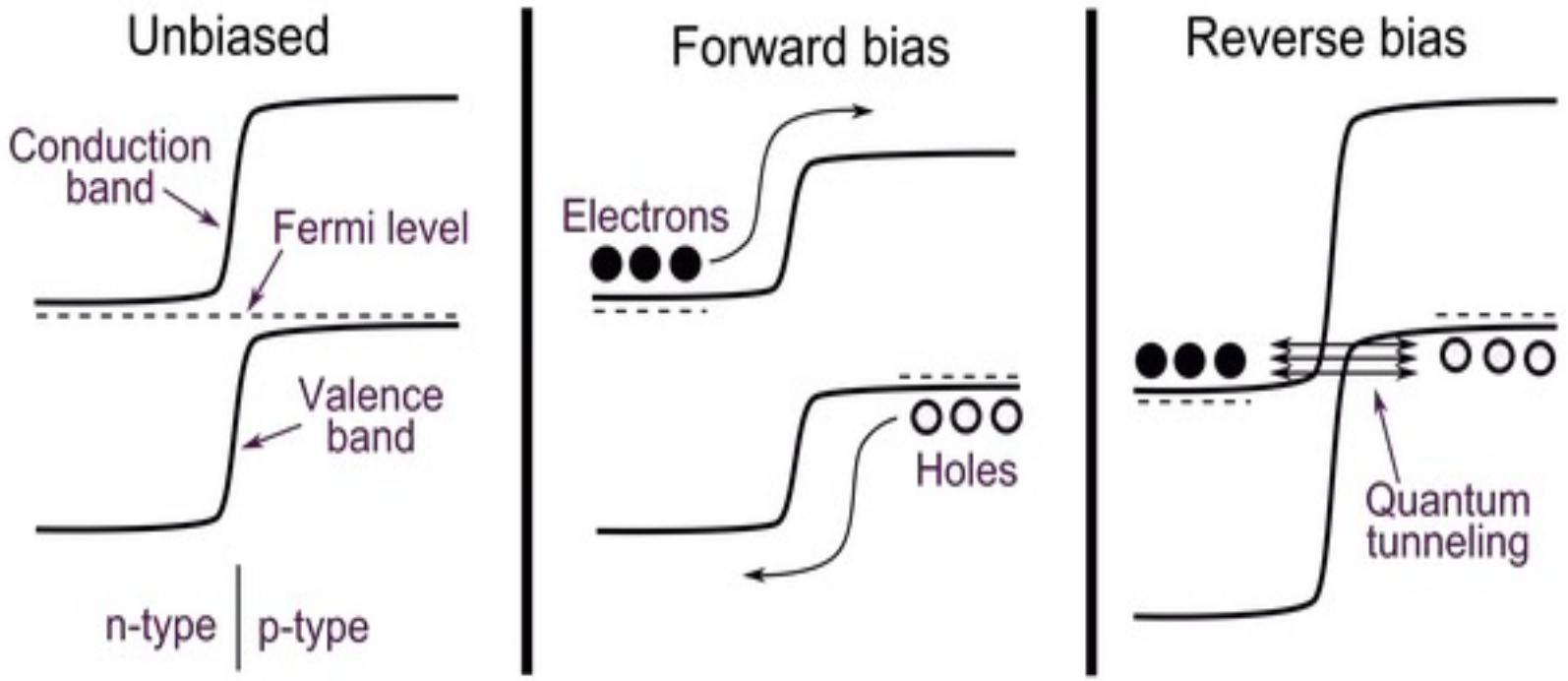


The Biased n-type Band Diagram

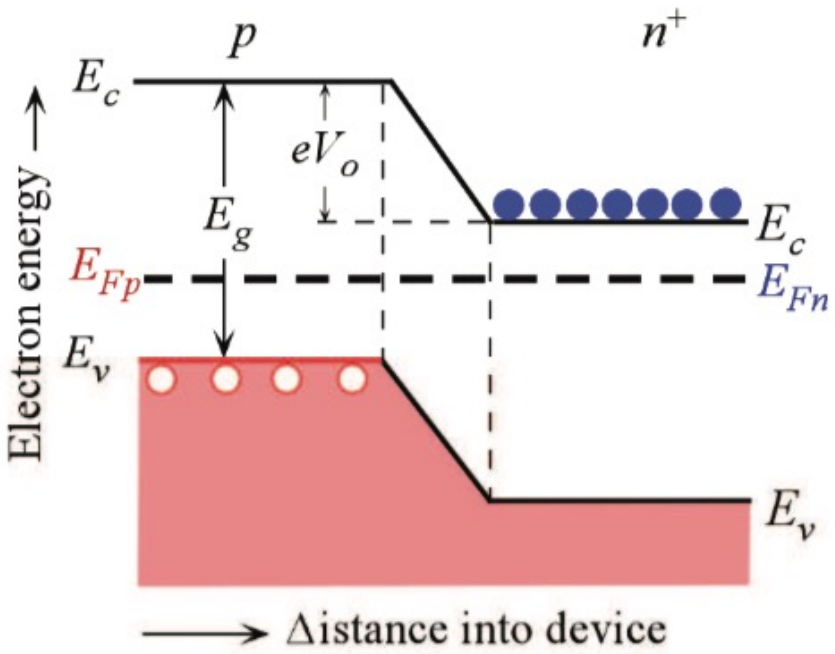


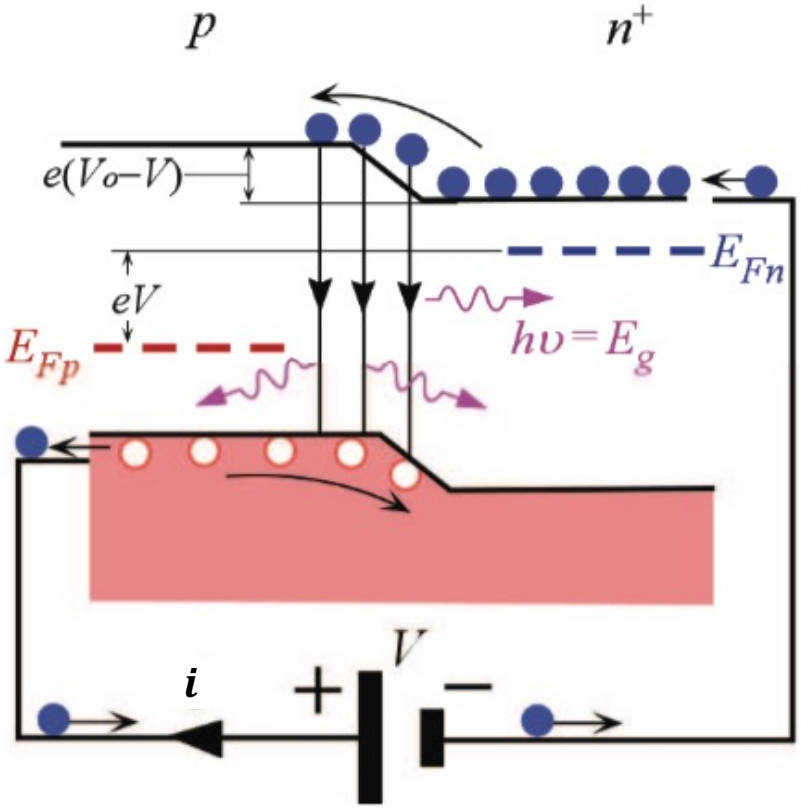


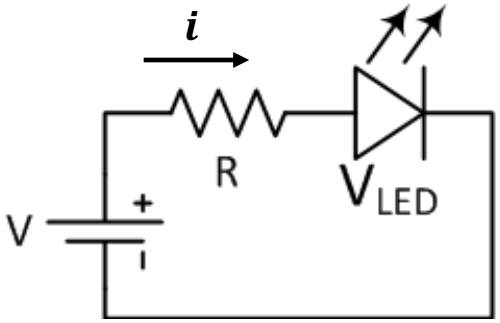
The Biased PN Junction Band Diagram



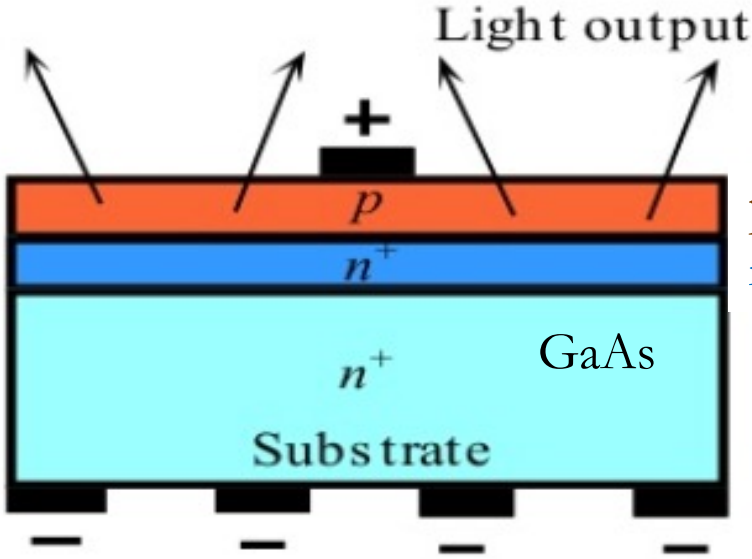
LED Basic Operation





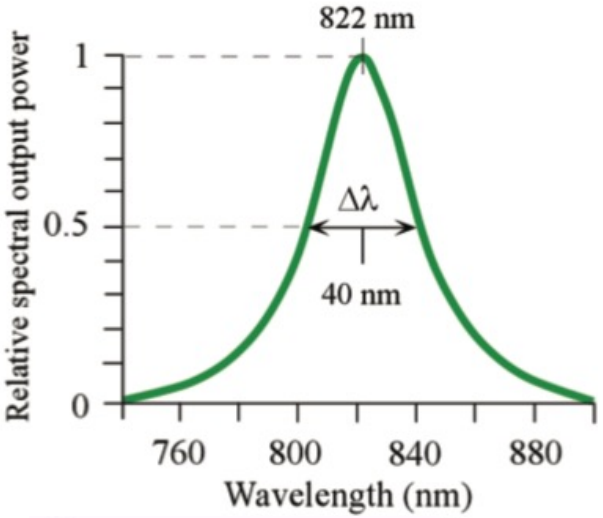


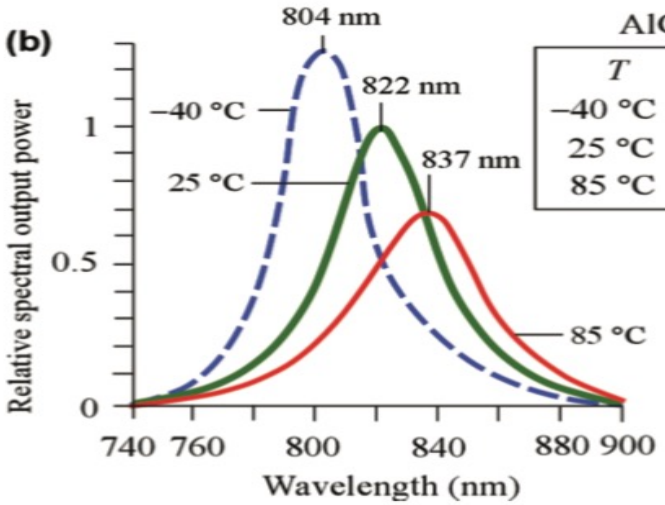
LED Basic Structure



p-AlGaAs

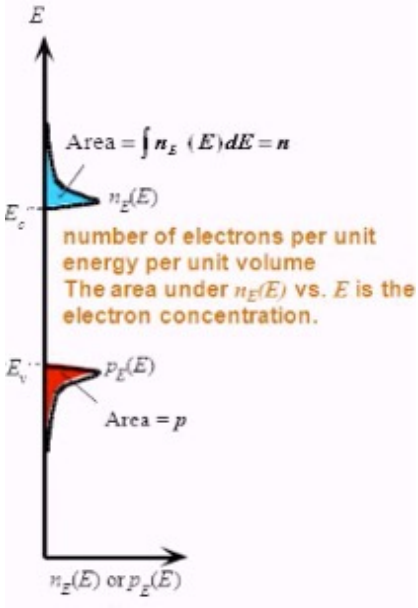
n-AlGaAs



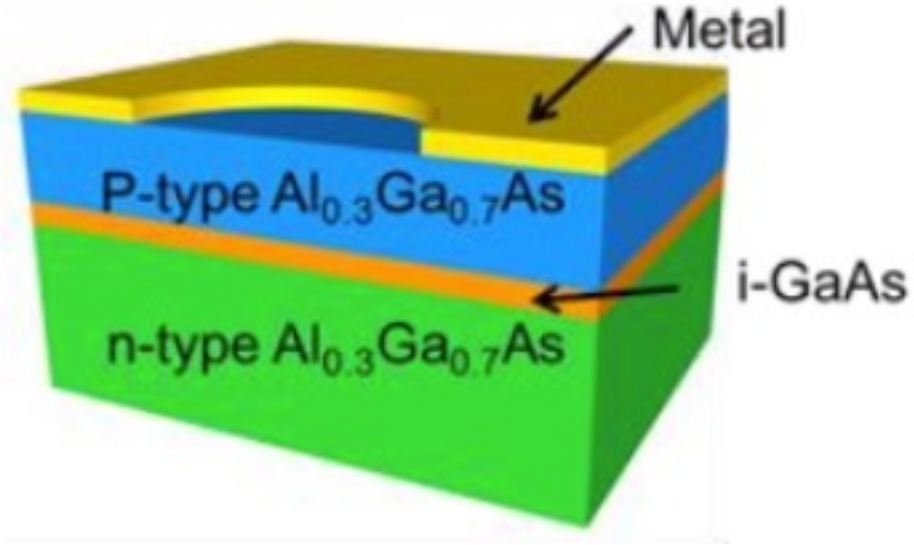


AlGaAs IR LED

T	λ_o	$\Delta\lambda$
−40 °C	804 nm	30 nm
25 °C	822 nm	40 nm
85 °C	837 nm	48 nm



LED Basic Structure



	Unit	GaAs	$Al_xGa_{1-x}As$, $0<x<0.45$
Bandgap Energy	eV	1.424	1.424 + 1.247x
Electron Effective Mass	m_0	0.067	0.067 + 0.083x
Hole Effective Mass	m_0	0.5	0.5 + 0.29x
Dielectric Constant	ε_0	13.1	13.1 – 3x
Conduction Band Discontinuity	%	-	$\Delta E_C \sim 67\% \Delta E_g$
Valence Band Discontinuity	%	-	$\Delta E_V \sim 33\% \Delta E_g$

Q & A

Send me an email if you have any questions.